

Hostel Maintenance & Issue Ticketing System

PROBLEM STATEMENT 07

Sanjeev Murthy – PES1UG24CS612

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Residents shall log maintenance tickets with room number, photo and issue severity.	High	Pass: Ticket created with ID and routed to warden. Fail: Missing room number accepted.	Core reporting function.
FR-002	Functional	Warden shall assign tickets to maintenance staff.	High	Pass: Assignment recorded. Fail: Ticket remains unassigned.	Dispatches issues.
FR-003	Functional	Maintenance staff shall update assigned ticket status.	High	Pass: Status changes to In Progress/Resolved. Fail: Change not recorded.	Tracks progress.
FR-004	Functional	System shall track each ticket's SLA timeline.	High	Pass: SLA status reflects recorded time. Fail: SLA not updated.	Monitors delays.
FR-005	Functional	Residents shall provide closure sign-off for resolved tickets.	Medium	Pass: Sign-off recorded. Fail: Ticket closes without sign-off.	Confirms resolution.
NFR-001	Nonfunc.	System shall send SMS/Email escalation if high-priority tickets exceed 24-hour SLA.	High	Pass: Alert is triggered during SLA test.	Escalates overdue issues.
NFR-002	Nonfunc.	System shall restrict ticket access and modification to authorized users.	High	Pass: Unauthorized access is denied.	Protects ticket data.

ACTORS:

- Hostel Resident — logs tickets and gives closure sign-off.
- Maintenance Warden — assigns tickets and tracks SLA.
- Maintenance Staff — updates ticket status.

Use Cases

1. Log Maintenance Ticket — Resident reports a maintenance issue.
2. Attach Photo — Photo is added to the ticket.
3. Assign Maintenance Staff — Warden assigns the ticket.
4. Update Ticket Status — Staff updates progress/resolution.
5. Track SLA Timeline — System tracks the ticket SLA.
6. Closure Sign-off — Resident confirms resolution.
7. Send SLA Escalation Alert — System alerts on overdue high-priority tickets.

USE CASES AND ACTORS

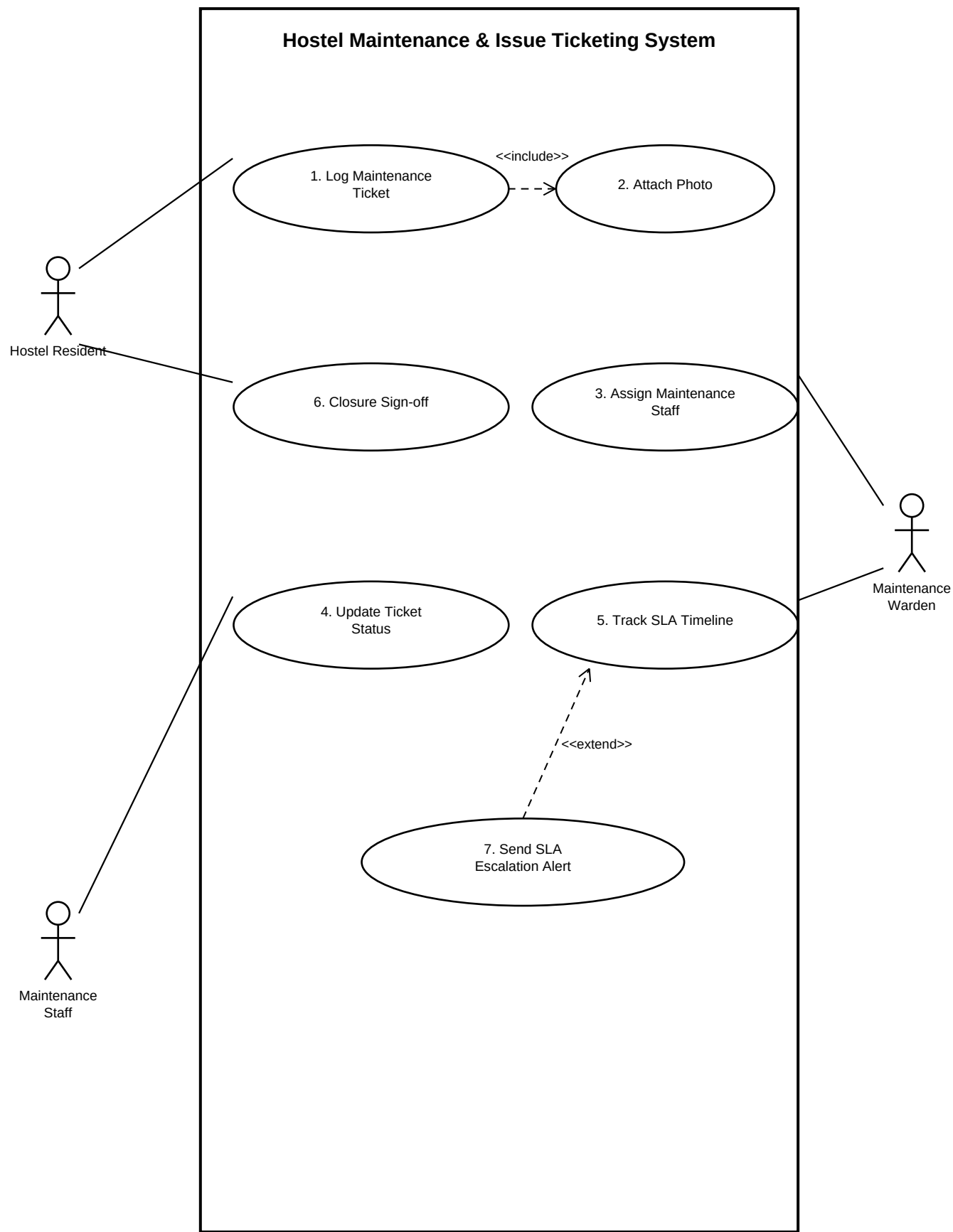
Hostel Resident: 1) Log Maintenance Ticket 2) Closure Sign-off

Maintenance Warden: 1) Assign Maintenance Staff 2) Track SLA Timeline

Maintenance Staff: 1) Update Ticket Status

UML DIAGRAM

Hostel Maintenance & Issue Ticketing System



Solid slant lines = actor associations Dashed arrows = UML relationships

USE CASE FLOWCHART

Use-Case Flow Specification – Log Maintenance Ticket

Use Case: Log Maintenance Ticket

Actor: Hostel Resident

Preconditions

1. Resident is authorized.
2. Resident has a valid room number.
3. A maintenance issue has been identified.

Postconditions

1. Ticket is recorded with a unique ID.
2. Ticket is routed to the Warden queue.
3. Ticket details are stored.

Main Success Scenario

1. Resident opens the maintenance ticket option.
2. System displays the ticket form.
3. Resident enters room number and issue details.
4. Resident selects severity and attaches a photo.
5. System validates the information.
6. System creates the ticket and unique ID.
7. System routes the ticket to the Warden queue.
8. System displays confirmation.

Alternate Flow – Missing Room Number

- 4a. Resident submits without room number.
- 4b. System rejects the ticket and requests the room number.
- 4c. Resident enters the room number and resubmits.
- 4d. Flow continues from Step 5.